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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

Name of Student:	
Date of Performance	**Date of Completion**

Experiment No. 1

Experiment Name: Develop a Complete Web Page Using HTML Basic Tags

Aim/Objective

The primary goal of this experiment is to **understand and apply fundamental HTML tags** to construct a simple, well-structured web page. You will learn to incorporate essential elements such as text formatting, lists, paragraphs, tables, and images.

Pre-requisites

- A text editor (e.g., VS Code, Notepad++, Sublime Text).
- A web browser (e.g., Chrome, Firefox, Edge).

Theory/Background

HTML, or HyperText Markup Language, is the foundational language for creating web pages. It uses a system of elements, defined by **tags**, to structure and give meaning to content. The browser interprets these tags to render the page as intended.

A basic HTML document structure consists of the following key tags:

- <!DOCTYPE html>: Declares the document as HTML5.
- <html>: The root element that contains all the page's content.
- <head>: Contains meta-information about the page, such as the title. This content is not displayed on the page itself.
- <title>: Sets the title that appears in the browser's tab or window title bar.
- <body>: Contains all the visible content of the web page, including text, images, and other elements.
- <h1> **to** <h6>: Defines headings of different levels. <h1> is the most important.
- <p>: Defines a paragraph of text.
- <ul> **and** <ol>: Create unordered (bulleted) and ordered (numbered) lists, respectively. Each item in a list is defined by an <li> tag.
- <table>: Structures data in rows and columns. It contains <tr> (table row), <th> (table header), and <td> (table data) tags.
- <img>: Embeds an image. It is a self-closing tag and requires the src (source) and alt (alternative text) attributes.

Task/Problem Statement

Create an HTML file named my_first_page.html that contains the following elements:

1. A document title: "My First HTML Page".
2. A main heading (<h1>).
3. An introductory paragraph (<p>).
4. Demonstrations of text formatting tags: **bold (<b>), italic (<i>), and underline (<u>)**.
5. An unordered list of your favorite hobbies.
6. An ordered list of your top 3 programming languages.
7. A simple table with two columns, displaying two subjects and their corresponding marks.
8. An image.

Procedure/Steps

1. **Open a text editor** and create a new file.
2. **Add the basic HTML structure** to the file.

```
<!DOCTYPE html>
<html>
<head>
  <title>My First HTML Page</title>
</head>
<body>
  <!-- All content goes here -->
</body>
</html>
```

3. **Insert the main heading** and an introductory paragraph inside the <body> tags.

```
<h1>Hello, World! Welcome to My Page</h1>
<p>This is my first attempt at building a web page using basic HTML. I'm excited to learn more about web development.</p>
```

4. **Add the text formatting example** within a new paragraph.

```
<p>This text is <b>bold</b>, this is <i>italic</i>, and this is <u>underlined</u>.</p>
```

5. **Create the unordered list** of hobbies.

```
<h2>My Hobbies</h2>
<ul>
  <li>Reading</li>
  <li>Playing sports</li>
  <li>Coding</li>
</ul>
```

6. **Create the ordered list** of programming languages.

```
<h2>Top 3 Programming Languages</h2>
<ol>
  <li>Python</li>
  <li>JavaScript</li>
  <li>Java</li>
</ol>
```

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7. **Build the simple table** for subjects and marks.

```
<h2>Subject Marks</h2>
<table border="1">
  <tr>
    <th>Subject</th>
    <th>Marks</th>
  </tr>
  <tr>
    <td>Web Programming</td>
    <td>95</td>
  </tr>
  <tr>
    <td>Data Structures</td>
    <td>90</td>
  </tr>
</table>
```

8. **Add an image.** Make sure to place an image file (e.g., lab_image.jpg) in the **same directory** as your HTML file.

```
<h2>An Image</h2>
<img src="lab_image.jpg" alt="Description of the image" width="300" height="200">
```

9. **Save the file** as my_first_page.html.
10. **Open the file in your browser** to view the result.

Expected Output

The final web page should display the heading and paragraphs, the formatted text, a bulleted list of hobbies, a numbered list of programming languages, a bordered table with subject marks, and the embedded image. The overall structure should be clear and readable.

Viva Voce Questions

1. What is the main difference between an ordered list and an unordered list?
2. Explain the purpose of the src and alt attributes in the <img> tag.
3. Why is the content inside the <head> tag not visible on the webpage?
4. What would happen if you forgot to include the <!DOCTYPE html> declaration?
5. Describe the structure of a basic HTML table.

Conclusion/Learning Outcome

By completing this experiment, you have successfully created a functional and structured web page using fundamental HTML tags. You now have a solid understanding of how to use headings, paragraphs, lists, tables, and images to organize and present content on the web. This is a crucial first step in your journey to becoming a web developer.

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